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Clinical Report Guided Retinal Microaneurysm Detection With Multi-Sieving Deep Learning

Ling Dai^{ID}, Ruogu Fang, Huating Li, Xuhong Hou, Bin Sheng^{ID}, Qiang Wu, and Weiping Jia

Abstract—Timely detection and treatment of microaneurysms is a critical step to prevent the development of vision-threatening eye diseases such as diabetic retinopathy. However, detecting microaneurysms in fundus images is a highly challenging task due to the low image contrast, misleading cues of other red lesions, and the large variation of imaging conditions. Existing methods tend to fail in face of the large intra-class variation and small inter-class variations for microaneurysm detection in fundus images. Recently, hybrid text/image mining computer-aided diagnosis systems have emerged to offer a promise of bridging the semantic gap between images and diagnostic information. In this paper, we focus on developing an interleaved deep mining technique to cope intelligently with the unbalanced microaneurysm detection problem. Specifically, we present a clinical report guided multi-sieving convolutional neural network, which leverages a small amount of supervised information in clinical reports to identify the potential microaneurysm regions via the image-to-text mapping in the feature space. These potential microaneurysm regions are then interleaved with fundus image information for multi-sieving deep mining in a highly unbalanced classification problem. Critically, the clinical reports are employed to bridge the semantic gap between low-level image features and high-level diagnostic information. We build an efficient microaneurysm detection framework based on the hybrid

text/image interleaving and validate its performance on challenging clinical data sets acquired from diabetic retinopathy patients. Extensive evaluations are carried out in terms of fundus detection and classification. Experimental results show that our framework achieves 99.7% precision and 87.8% recall, comparing favorably with the state-of-the-art algorithms. Integration of expert domain knowledge and image information demonstrates the feasibility of reducing the difficulty of training classifiers under extremely unbalanced data distributions.

Index Terms—Diabetic retinopathy, fundus image analysis, multi-sieving CNN, microaneurysm detection, clinical reports, deep learning.

I. INTRODUCTION

DIABETIC Retinopathy (DR) is the leading cause of blindness globally and also the top reason of new cases of legal blindness among adults aged 20 to 74 years in the United States [1]. Among an estimated 285 million people with diabetes mellitus worldwide, nearly one-third have signs of DR and among these, a further one third is vision-threatening [2]. DR-induced blindness is a personal catastrophe to the individual and it costs the United States approximately \$500 million annually [3]. Fortunately, the risk of vision loss caused by DR can be significantly reduced via early detection and timely treatment [4]–[7]. It is estimated that regular screening and timely treatment to preserve the vision of Americans at risk for developing DR yields a net annual saving to the government of more than \$100 million [8]. Micro-aneurysm (MA), the earliest clinical sign of DR, is defined as tiny aneurysms or swelling occurring secondary to the capillary wall and appear as small red dots in the superficial retinal layers, with the possibility of opening and leading blood into the retina tissue surrounding it. MA counts using stereoscopic color fundus photographs is an important measure of progression of retinopathy in the early stage and may serve as a surrogate endpoint for severe changes in some clinical trials [9]. However, manual segmentation and counting of MAs is time-consuming, subjective, error-prone and infeasible for large-scale fundus image analysis and diagnoses. Therein automatic detection and counting of MAs plays a key roles for early diagnosis and assessment of vision-threatening diseases and is, therefore, a core component of any computer-aided retinopathy diagnosis system.

However, several factors, including the variation in image lighting, the variability of image clarity, the occurrence of other red lesions, extremely low contrast, and highly variable

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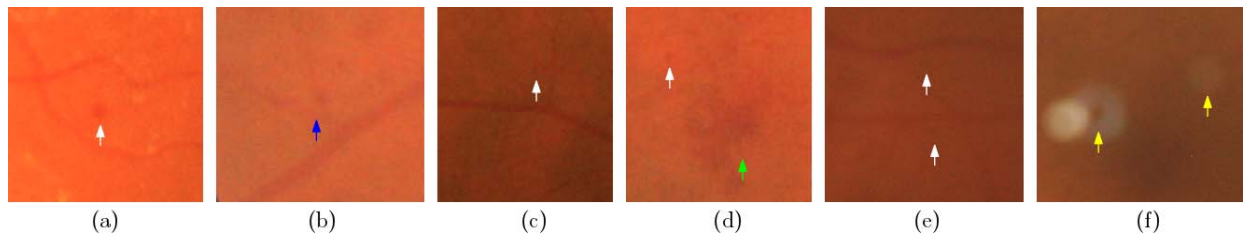


Fig. 1. Challenging cases of MA detection. (a) A normal and obvious microaneurysm (white arrow). (b) Blood vessel joint which appears similar to MA (blue arrow). (c) Light and texture of background vary even in the same fundus image. (d) Hemorrhage may influence detection of MA, causing false positive detection (green arrow points to hemorrhage and white points to MA). (e) The fundus image is blurred due to lens opacity, which makes detection of MA (white arrow) much more difficult. (f) Reflection (yellow arrow) in fundus images is the major noise in edge detection based methods.

image background texture, are usually considered as impenetrable barriers to the automatic MA detection for practical clinical usage. For example, normal MAs appear as small red dots in color fundus images, while small dot-shaped hemorrhage and vessels conjunctions may have the similar appearance, making the detection challenging for the filter-based and morphology-based method. Furthermore, most methods published would first generate candidates of MA and then remove those appears on vessels, connected with vessels or with a large area of hemorrhages because these candidates are most likely to be hemorrhages or part of vessels. But most microaneurysms are located near narrow vessels, which may be regarded as part of vessels and excluded by many existing methods in this prospect. Fig. 1 shows some examples of challenges caused of MA detection in fundus images as described above.

To address the above challenges, we propose a multi-modal framework utilizing both expert knowledge from text reports and color fundus images. Different from previous methods, our proposed framework is able to (1) integrate non-image information from experts through clinical reports; and (2) accommodate highly unbalanced classification in medical data problem. Specifically, we first train a weak image-to-text mapping model from text reports and the corresponding fundus images. This statistical model is used to generate a rough segmentation of lesions in fundus images. Then we train a convolutional-neural-network (CNN) using the clinical report guided information and the original fundus images as input to generate the final high quality segmentation.

The remainder of the paper is organized as follows: Section 2 introduces the related work on automatic MA detection. Section 3 presents the method followed by our experimental results in Section 4. Conclusions are drawn in Section 5.

II. RELATED WORK

Numerous approaches have been proposed for MA segmentation using retinal color fundus images. These approaches can be generally categorized into three groups including the approaches based on morphological operation, template matching, and supervised classification.

Early attempts on MA detection can be traced back to 1984, Lay *et al.* [10] developed a method based on morphological/thresholding and image enhancement techniques. They separated circular structures from elongated structures by

applying top-hat transformation with linear structuring element to green channel of the image. In this way, small, dark, dot-shaped MAs are distinguished from other dark tissue. Various modification [11]–[15] was done based on this primary morphology to increase the diagnose accuracy. While this kind of hand-crafted image processing is usually fast and easy to implement, the capability of the method is limited by the designer. Some important hidden structures and unnoticeable patterns may be ignored by the designer and cause false segmentation.

The statistical result shows that the intensity distribution of MAs is similar to Gaussian distribution [16]. Based on this fact, template matching based MA detection methods were proposed and greatly improved the segmentation accuracy. Quellec *et al.* [16] proposed a wavelet transform based matching to detect MAs. In this method, MAs were detected using local template matching in the wavelet domain. Zhang *et al.* [17] proposed a multi-scale correlation coefficient based method. In this method, MA candidates were detected by applying a non-linear filter with five Gaussian kernels at different standard deviations to the input image. Similar to the morphological method, this kind of methods still suffer from ignorance of hidden structures and unnoticeable patterns. And usually, a lot of parameters need to be determined empirically.

With the flourish of machine learning, publications on MA detection using classification based methods are frequently seen recently. Antal and Hajdu [18] proposed a rule-based expert system to detect MAs. In this method, MA candidates are first selected from fundus images, and then a rule-based classifier is used to find true MAs. Niemeijer *et al.* [19] presented a hybrid scheme using both top-hat based method and a supervised classification. In this method, red lesion candidates were first detected and then a classifier is trained to distinguish MAs from non MAs. These machine learning based methods, although succeed in finding hidden patterns of features and MAs, still rely on manually selected features and empirically determined parameters.

To solve these problems Gulshan *et al.* [20] proposed a deep learning based method to automatically grade DR in fundus images. In this method, deep neural network is used to process raw images directly and output the grading result of DR. While this work successfully solved the problem of finding hidden patterns and empirically determined parameters is not need, it requires large amounts of fundus images and their annotations for training the classifiers, which is costly

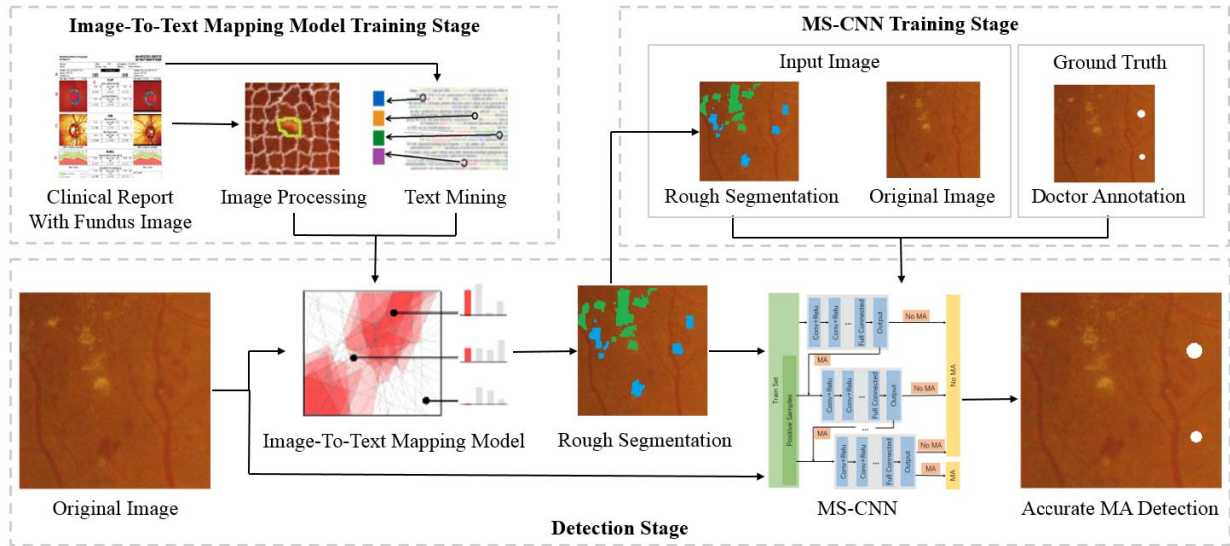


Fig. 2. Overview of the proposed clinical report guided multi-sieving convolutional neural network (MS-CNN) framework. Original fundus images are first roughly segmented using the image-to-text mapping model jointly learned from the clinical reports and the fundus images. This segmentation result is coded as an extra semantic channel and feed into MS-CNN with the original fundus image for precise MA detection.

and time-consuming. Besides, there is no quantitative results generated explicitly for a specific lesion. These quantitative data are of great importance in understanding the development of DR and monitoring its progression.

There are a number of work in semantic image segmentation [21], [22], and video captioning [23]–[25] related to our proposed work. Our approach is relevant to semantic image segmentation in the way that both methods classify the pixels in the image into several classes. And both video captioning and our method utilize similar classifiers such as CNN. However these three areas are fundamentally different in the following way: the main focus of semantic image segmentation is generally to segment large areas of objects; video captioning focuses on classifying a whole image or video into several classes. As for our method, we try to detect tiny lesions in fundus images. If the semantic segmentation approaches are directly applied to our use case of MA detection, the methods tend to fail in identifying tiny objects, leading to low precision and recall. We show the results of detecting MA using fully convolutional networks [26] in Section IV. On the other hand, our detection result contains richer information of the tiny lesions, including the positions of each MA, which is missing in video captioning.

Shin *et al.* [27] also published interesting work on interleaved text/image deep mining. This work extracts and mines the semantic interactions between radiology images and clinical text reports. This system interleaves between unsupervised learning and supervised learning on the document and sentence-level text collections. Given an image of a patient scan, semantic topics in radiology images are predicted, and associated keywords are generated. The goal of our work is different from this paper While Shin's work generates an image-level prediction based on the analysis of the whole image, our method generated a superpixel-level prediction of lesion types. During the text mining stage, Shin's work

generates topics by mining the text reports, while we focus on counting the occurrences of lesion types using clinically relevant keywords. Furthermore, we use the image-to-text mapping as an extra channel to the color images to feed into multi-sieving CNN for the classification, which is not included in Shin's work as they stop at the image-level keywords generation.

Seoud *et al.*'s [28] work presented a novel method for automatic detection of both microaneurysms and hemorrhages in color fundus images. The main focus of this work is a new set of shape features, called Dynamic Shape Features, that do not require precise segmentation of red lesion regions. Their method detects all types of red lesions in fundus images, such as MA and hemorrhage, without distinguishing between them. While MAs are early signs of DR, hemorrhages can already affect eyesight. Distinguishing between these lesion types is very important in the clinical practice. Furthermore, MA count is a critical parameter that classifies fundus images into None DR, light DR, or worse DR, while hemorrhage is a critical marker to differentiate between mild and severe DR. Mixing the detection of MA and hemorrhage is not a practical solution in the medical field. In comparison, our work focuses on detecting MAs in fundus images, which is clinically important for DR grading.

III. METHODOLOGY

The overview of our proposed approach is illustrated in Fig. 2. It includes two parts: an automatic image-to-text mapping that generates expert guided segmentation from clinical reports coupled with fundus images, and a multi-sieve convolutional neural network (MS-CNN) that filters the false positives based on multimodal inputs including the expert-guided segmentation and the image data. Specifically, in the training stage, the image-to-text mapping model extracts key words from expert annotations in clinical reports to establish

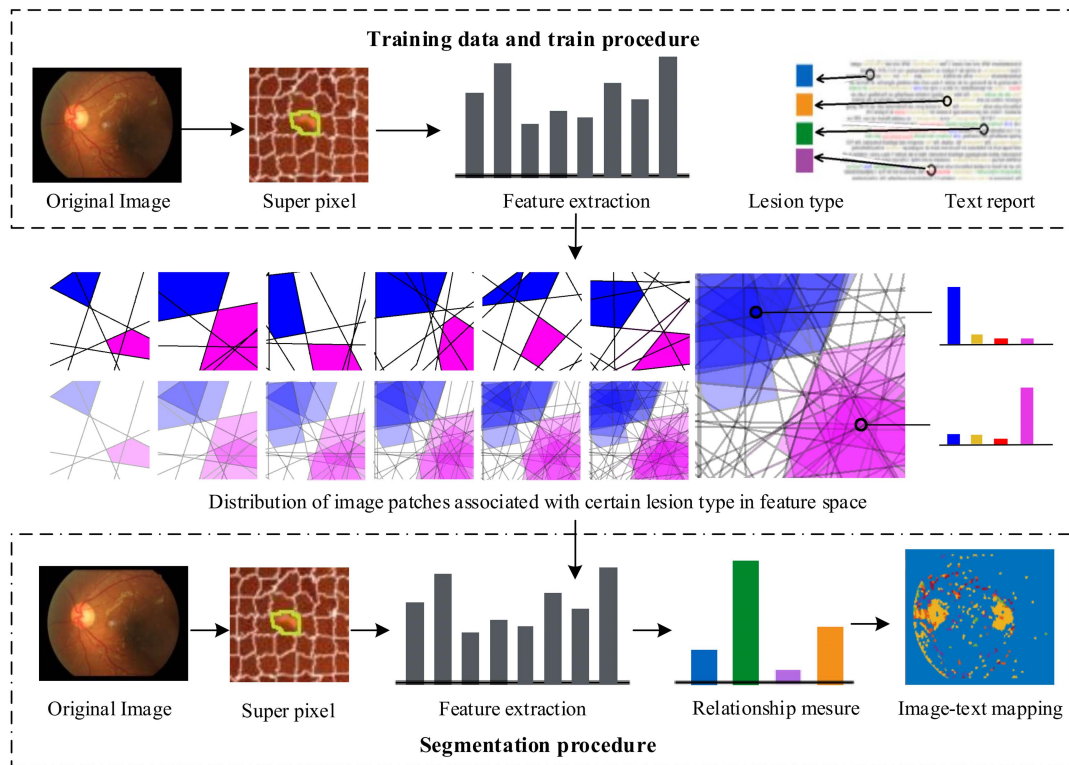


Fig. 3. Image-to-text mapping model illustration. The top box illustrates the training procedure of the image-to-text mapping model, where each fundus image is segmented into superpixels which are mapped to the randomly partitioned feature space. A image-to-text mapping model is learned through the partitions' lesion label distribution (middle row). At the test time, the images are segmented into super-pixels which are also mapped to the feature space for lesion label prediction (bottom row).

a mapping between keywords and visual feature subspaces (Fig. 3). This model maps visual features to lesion types projected from the text, which leads to expert-guided segmentation of the fundus images (Section III-A). After that convolutional neural networks are learned from multimodal sources (the expert-guided segmentation and the fundus images) since it can automatically learn the feature maps and perform end-to-end pixel-wise classification. However, different from classifying balanced datasets, classifying MAs in fundus images is more challenging, due to the highly unbalanced data with only around 0.1% of pixels being MAs in an entire fundus image. Therefore, traditional CNN methods result in a large number of false positives. Furthermore, it is important to integrate multiple information sources from clinical reports and fundus images, so the available information from both sources can be thoroughly explored. Our approach is a multi-sieve filtering scheme that stresses challenging false positives in the cascaded CNN architecture, and the integration of multiple cues from fundus images and clinical reports (Section III-B). At the test time, the learned image-to-text mapping model in the training stage is used to gain a rough segmentation from the new test fundus image, without the need of clinical reports at this stage since the expert guidance is already embedded in the image-to-text mapping model. Then the rough segmentation is fed into the cascaded MS-CNN with the color fundus images as the fourth layer for real-time pixel-wise classification. In the rest of the paper, we introduce the details of the proposed

expert-guided learning and the multi-sieving strategy for unbalanced MA detection in fundus images.

A. Learning Image-to-Text Mapping Model

In this section, we introduce the image-to-text mapping model to extract expert knowledge from clinical reports. In the training stage, the input to the model is fundus images with their corresponding clinical text reports, and the output is a probabilistic map of the lesion types in fundus images. In the testing stage, the input of the model is a fundus image only, and the output is a probabilistic map of the lesion types in this image. Our proposed image-to-text mapping model consists of five stages, as shown in Fig. 3: (1) image preprocessing and feature extraction; (2) text information extraction from clinical reports; (3) random feature space partition; (4) lesion distribution model; (5) mapping features to lesion types.

1) Image Preprocessing and Feature Extraction: To eliminate the large variation of imaging conditions of fundus images such as lighting and contrast, a pipeline of preprocessing steps are performed. First, all images are resized to the same image size (i.e. 1200 x 1800 pixels). Then histogram balance is performed to balance the intensity difference across images. To improve the computational efficiency, the fundus images are over-segmented into super-pixels based on the similarity between neighboring pixels using Simple Linear Iterative Clustering (SLIC) [29], where the pixels belonging to the same

TABLE I

AN EXAMPLE OF CLINICAL TEXT REPORTS AND THE LESION-RELATED INFORMATION EXTRACTED. THE BOLD WORDS IN THE REPORT CONTENT ARE KEYWORDS OF DIFFERENT KINDS OF LESIONS, THE BOLD UNDERLINED WORDS ARE NEGATIVE WORDS AND THE BOLD ITALICIZED WORDS ARE POSITIVE WORDS. AND THEN THE TEXT REPORT IS TRANSFORMED INTO A BINARY ARRAY INDICATING WHETHER A CERTAIN KIND OF LESION APPEARS IN IMAGES, SHOWN IN RIGHT COLUMNS. MA, HE, SE, HH REPRESENTS MICROANEURYSM, HARD EXUDATE, SOFT EXUDATE, AND HEMORRHAGE RESPECTIVELY

Clinical Reports	MA	HE	SE	HH
There <i>exist</i> microaneurysm and dot hemorrhages in posterior pole, <i>probable hard exudate</i> at upper temporal.	1	1	0	0
There <i>exist</i> microaneurysm and dot hemorrhages at posterior pole, hard Exudate above macular in the shape of a circle.	1	1	0	0
Microaneurysm and dot hemorrhages scattered in posterior pole, line shaped hemorrhages are <i>seen</i> near vascular arcade, soft exudate are seen in upper temporal and temporal side of optic nerve head.	1	0	1	1
Microaneurysms are seen below the posterior pole, no sign of hemorrhages, no hard exudates. Suspected soft exudates exists near vascular arcade.	1	0	0	0
Exudates and hemorrhages are not seen in the field. Microaneurysms are not clear .	0	0	0	0
Exudates are absent in the field. Microaneurysms are not clear .	0	0	0	0
There are no signs of DR .	0	0	0	0

super-pixel are assumed to have the same label (e.g. lesion types or normal). Features are further extracted from super-pixels using a pre-trained CNN model, where we use the fully connected neural network layer of AlexNet [30] which has a high representation power of image features. The choice of the fully connected layer is motivated by the small output feature dimensionality from this layer, where the high-dimensional features from the convolutional layers may suffer from the curse of dimensionality. We also choose a rectangle centered at the center of superpixels as the input of CNN because the shape of superpixels is not regular.

2) *Text Information Extraction*: In this step, we extract the lesion information from clinical text reports written in natural languages. For each sentence in the clinical report, we attempt to determine (1) which kind of lesion is described; (2) whether the lesion is present or absent; (3) where is the lesion roughly located.

Typically biomedical data mining algorithms fall into three categories: dictionary-based approaches [31]–[33], rule-based approaches [34]–[36], and machine learning approaches [37]. All of the three methods have their own advantages and disadvantages [38]. The observation behind our approach is that clinical text reports always contain lesion names and rough locations of the lesion mentioned. Unlike natural languages in the wild, texts in clinical reports are usually structured in certain formats. We extract information from clinical reports by combining rule-based methods and dictionary-based methods.

We first establish a word dictionary of the lesions of interest such as “microaneurysm”, “hemorrhage”, and “hard exudate”. Then we establish another word dictionary of the descriptive words such as “no”, “not”, “none”, “slightly”, “present”, “absent”, each of them are classified as positive (such as “present”) or negative words (such as “absent”, “no”). For each sentence in the clinical reports, we first match each word with the lesion dictionary to find the lesion type described in the sentence. Then we calculate whether the lesion described are absent or present using the following rules: (a) The initial status of lesion is “present”; (b) if we find a positive word in the descriptive word dictionary, then

keep the current status; (c) if we find a negative word in the descriptive word dictionary, we toggle the status (change present to absent or absent to present). Following the rules described above, we can transcript the natural language in text reports into binary vectors, with 1/0 representing the presence/absence of a specific lesion. This straightforward and effective technique enables us to efficiently extract the key information from clinical reports while avoiding impractical manual parsing used in methods such as semantic segmentation [26] which are laborious and time-consuming. In Table I, we show examples of text information extracted from clinical text reports. We perform a word-level search from clinical reports, as shown in the first column of Table I, to find lesion keywords such as “microaneurysm”, “hemorrhage”, and “hard exudate” and the descriptive words such as “no”, “absent”, and “slightly”. Then the matched lesion type will be coded in as binary indicators 1 or 0 for the corresponding lesion type based on the descriptive words. With this approach, the original text reports are encoded into binary vectors.

3) *Random Feature Space Partitioning*: This section introduces the feature space partitioning approach used in our proposed framework. To find the optimal feature combinations for each lesion type, we first randomly partition the feature space into subspaces, each of which represents a different feature combination. We then search for the most representative subspace for each lesion type. The detailed approach is described below.

Given the image visual features and the corresponding lesion types from the previous steps, we map the lesion types as probabilities to the partitions in the feature space. The intuition for this step is due to the missing location information of the lesions in the text reports. Therein we use a semantic mapping from visual features [39] to fill in this gap. To this end, the feature space is first randomly and uniformly partitioned using the random fern approach [40] which generates an index value for each subspace. A fern is a sequence of binary decision functions $\mathcal{F}$. These functions map the feature vectors of super-pixels to integer indices of the partitioned space. Suppose there are L binary functions

in $\mathcal{F}$, as each binary function $\mathcal{L}_i$ ($i = 1, \dots, L$) produces a bit in $\mathcal{F}$, the resultant binary code represents values in the range 1 to 2^L . The probability distribution of lesion types in each feature space partition is computed by counting the number of super-pixels with each lesion type. The mapping between the super-pixels and lesion types is obtained by assigning the lesion types in the clinical reports to all super-pixels in the corresponding fundus image. However, this counting method tends to prioritize lesion types that are common for all partitions and therefore suffer from less discriminative power. To overcome this challenge, we propose a new lesion distribution model as described in the next step.

4) Lesion Distribution Model: Inspired by the “term frequency-inverse document frequency” (*tf-idf*) model in natural language processing [39], we developed a model called “partition frequency-inverse lesion frequency” (*pf-ilf* model) to identify the best feature partitions to represent each lesion type.

The standard *tf-idf* model is well known in information retrieval, where term frequency $tf(t, d)$ is the raw frequency of term t divided by the maximum frequency of any term in the document d .

$$tf(t, d) = \frac{f_{t,d}}{\max_{t' \in d} f_{t',d}} \quad (1)$$

Inverse document frequency is defined as the logarithm of the inversion of the document frequency, which is the number of documents containing the term t , n_t , divided by the total number of documents N .

$$idf(t) = \log_2 \frac{N}{n_t} \quad (2)$$

Suppose there are N documents, and n_i documents contain word i . $f_{i,j}$ is the number of occurrences of word i in document j . The score of word i in document j is defined as the product of tf_{ij} and idf_i

$$tf.idf(t, d) = tf(t, d) \cdot idf(t) \quad (3)$$

In analogy, the two parts of our proposed “partition-frequency-inverse lesion frequency” (*pf-ilf*) model are used to measure two important properties of a feature space partition: uniqueness and importance, respectively. The “partition frequency” *pf* score measures the importance of each partition. If one feature space partition appears frequently in a lesion, it may carry important information in that lesion. Unfortunately, there exist some meaningless feature partitions (i.e. background regions that appear in every lesion type) that appear frequently in a majority of lesions. So we need the “inverse lesion frequency” (*ilf*) part to eliminate these feature-space partitions. According to the definition, the *ilf* score of a partition becomes larger when the number of lesions containing the partition becomes smaller. So if a partition has high *pf-ilf* score, this partition appears frequently in a lesion type and it appears in only a small group of lesions.

To obtain the *tf-ilf* score, we first find the lesion types in one image by identifying the keywords from text reports. Then we segment the image into super-pixels, map them into feature space, which are subsequently randomly partitioned. Our goal

is to find which partitions of the feature space can best model certain lesion types. Assume that a partition p in feature space models lesion l well, p will have two properties: 1) most super-pixels containing lesion l will be mapped into p , 2) few super-pixels with other lesions will be mapped into p . These two properties are similar to terms in documents in the original *tf-idf* model. As an analogue to *tf-idf* model, we define partition-frequency as

$$pf(p, l) = \frac{1 + f_{p,l}}{L + \max_k f_{k,l}} \quad (4)$$

and we define the inverse lesion frequency as

$$ilf(p) = \log_2 \frac{L}{n_p} \quad (5)$$

where $f_{p,l}$ is the number of the super-pixels with lesion label l that fall in the feature space partitions p . L is the total number of lesion types, n_p is the number of lesions that fall into partition p . Laplacian smoothing is applied to *pf* to avoid zero values for rare lesion types.

All together, we can define the score *pf-ilf* of a partition p for a lesion type l as

$$pf-ilf(p, l) = \frac{1 + f_{p,l}}{L + \max_k f_{k,l}} \times \log_2 \frac{L}{n_p} \quad (6)$$

With the proposed lesion distribution model, we can find the feature space partitions p that best models a specific lesion type l , by ranking the *pf-ilf* scores of all partitions for lesion type l . The middle part of Fig. 3 visualizes the random subspace partitioning of the visual feature space and the mapping between feature space partitions to lesion types.

In summary, if certain partition p in the feature space can represent a certain lesion l , most superpixels with lesion l will be mapped to p , thus the *pf* score must be high for p . And there are few super-pixels with lesions other than l that are mapped to p , thus the *ilf* score must also be high for p . In this way, we can find the most representative segmentations for a certain kind of lesion by searching for the segmentation with highest *pf-ilf* score.

5) Mapping Visual Features to Lesion Types: This last step in the image-to-text mapping model is to predict lesion types for each super-pixel in the test images, as illustration in the middle and bottom rows of Fig. 3. For each lesion type l , we pick the top k partitions with highest *pf-ilf* defined as $P_l(k)$. Suppose a super-pixel s is mapped to a set of partitions Q_s in the feature space. We define a final score of the super-pixel s and the lesion type l as $S(s, l) = |P_l(k) \cap Q_s|$, indicating the size of the intersection set between $P_l(k)$ and $Q(s)$. Finally the super-pixel s is labeled with the lesion type l with the highest score $S(s, l)$.

B. Multi-Sieving Convolutional Neural Networks for MA Detection

To determine whether a pixel is MA or not, a 64×64 patch around the target pixel is fed into our proposed MS-CNN classifier. Specifically, if we want to determine whether a pixel at (x, y) is MA, we take the patch at $(x - \frac{r}{2} : x + \frac{r}{2}, y - \frac{r}{2} : y + \frac{r}{2})$ as the input of MS-CNN, where $r = 64$ is the size

of the patch. Then CNN classifies whether the given pixel is MA based on the $r \times r$ patch around the pixel. We choose the size 64 because it is larger than the size of MAs, and the computation cost is generally acceptable. Rather than classifying each pixel, which is time-consuming and unnecessary, we first search for potential candidates in the image using the candidate selection method described in Fleming *et al.*'s work [41]. We first use a mean filter to the green channel of the original image. Then we discriminate MAs from blood vessels using the top-hat transform and a Gaussian shaped filter with size 21×21 . After that we threshold the output image and get MA candidates. We used the training dataset to search for the optimal threshold value in the final step. We adjust the threshold such that the candidate selection algorithm captures all MAs in the fundus image while keeping the number of candidates as small as possible. In our experiment, we get an average of 3600 MA candidates per image. Then we use MS-CNN to determine whether the candidates are MAs or not. Since the detected pixels may be connected with each other, we take the gravity center of each connected component as the center of an MA. For each detected center of MA, if its position is within 5 pixels to an MA annotated by doctors, we regard it as a successful detection. Otherwise, the detection is considered as false positive. On the other hand, if there are no pixels detected as MA within the radius of 5 pixels to an annotated MA, this MA is considered as a false negative.

In spite of its efficacy in large-scale image segmentation and object detection [30], CNN still faces limitations when dealing with the use case in this paper – detecting the MAs in fundus images. First of all, it favors balanced datasets, while the ratio of positive examples (MAs) over the negative examples (non-MAs) can be as low as 1:1000. Second, multiple misleading visual traits such as lighting, blood vessels (as described in Section I) can lead to erroneous classification using only visual features of the fundus image. In other words, non-image information would provide critical meta-data to guide the classification model by integrating additional cues such as expert knowledge from the clinical reports. The right image in the bottom of Fig. 3 visualizes the additional input information from the image-to-text mapping model. The integration of image and non-image information provides complementary cues for challenging regions.

To address the unbalanced dataset issue, we are the first to propose Multi-Sieving CNN (MS-CNN) to address the highly unbalanced dataset. As shown in Fig. 4, MS-CNN is a cascaded CNN model with the false positives from the previous network fed into the next network as negative examples. In the training phase, suppose all positive samples are in set P , and negative samples are in set N . For the first phase, we select all samples in P and randomly select an equal number of samples in N as initial training samples ($P^{(0)}$, $N^{(0)}$). Then for the n^{th} phase, we first perform classification using network trained in previous phase on all samples in $P^{(n-1)}$ and $N^{(n-1)}$. This will generate many false positive predictions, and collected in set $FP^{(n)}$. We select all positive samples in P again, but now randomly select an equal number of negative samples from $FP^{(n)}$, which is “hard” samples for the previous classifier. In the predicting phase, the inputs are fed into each

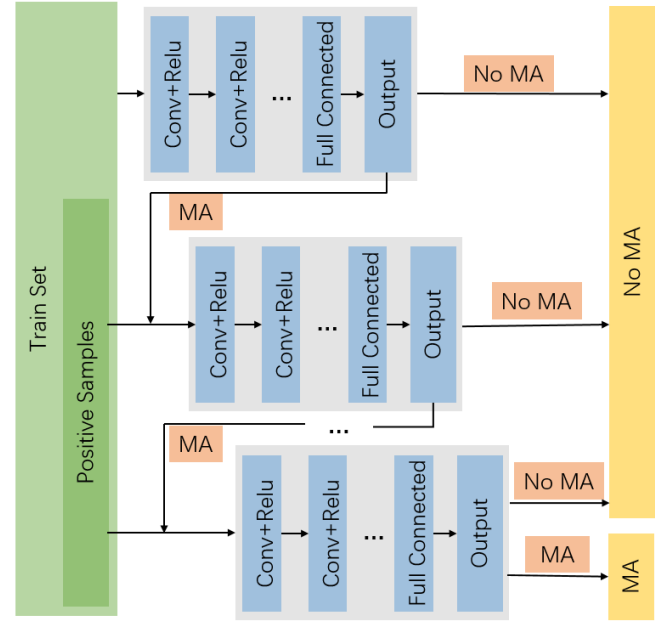


Fig. 4. The structure of MS-CNN. Each gray block is a CNN block, the green boxes are training samples, red boxes are classification output of each CNN block, yellow boxes are final classification result of MS-CNN. Each block in MS-CNN acts like filters, part of negative samples are eliminated in each block, leaving a smaller number of negative samples for next layer.

block of MS-CNN from top to bottom. Whenever a CNN block predicts the test data as negative, the overall MS-CNN prediction is regarded negative. Only if the prediction of all CNN blocks are positive, will the final classification result be positive. Algorithm 1 illustrates the multi-sieving training strategy.

Algorithm 1 MS-CNN Training Algorithm

Input: PositiveSamples: the set of positive samples
 NegativeSamples: the set of negative samples

- 1: $TrainingSet \leftarrow PositiveSamples \cup NegativeSamples$
- 2: **for** currentBlock $\leftarrow 1$ **to** NumBlocks **do**
- 3: $net[currentBlock] \leftarrow$ Pretrained CNN
- 4: Train $net[currentBlock]$ using $TrainingSet$
- 5: Perform Prediction On $PositiveSample \cup NegativeSamples$
- 6: **for** nSample $\leftarrow 1$ **to** $|TrainPositiveSample|$ **do**
- 7: Randomly select sample $s \in testFalsePositives$
 Add s to $TrainNegativeSamples$
- 8: **end for**
- 9: $TrainingSet \leftarrow TrainNegativeSamples \cup PositiveSamples$
- 10: **end for**
- 11: **return** net

The basic idea of the cascaded CNN classifiers is to train each stage of the CNN chain using different training data for different target samples. Each classifier in the cascaded CNN chain can be regarded as a filter that eliminates mostly the negative samples in its own stage. Then the following classifiers are trained without these negative samples that have been eliminated in the previous classifiers. That is, different

classifiers focuses on different kinds of samples. (e.g. the first classifier eliminates almost all of the obvious negative candidates that caused by image noise and hemorrhage, while small hemorrhage and MAs near vessels are harder to detect and many mistakes are made in these samples). We tune the classifier in the first stage to give a negative decision only when it is with high confidence. The hard samples are left for the later classifiers. The second stage then only focus on distinguishing MAs, small vessels, and small hemorrhages, which are much easier to train. To accelerate training and achieve better performance, we initiate the weights of CNN in each stage to be the same as the final weights of the previous stage rather than training from scratch. The weights will change due to the different training data fed into each classifier. Finally, we do not adopt the majority vote because it assigns the same weight to classifier, and fails to capture the different sensitivity of the CNNs in each stage.

The proposed multi-sieving-CNN (MS-CNN) share some commonality with boosting but they are distinctly different. Most boosting algorithms consist of iteratively learning weak classifiers with respect to a distribution and adding the weak classifiers together to form a strong classifier in the end. When the weak classifiers are summed together, they are typically weighted in some ways that are usually related to the weak learners' accuracy. The training data are then weighted as well so that the following classifiers can focus on the misclassified samples. While boosting tries to get a strong classifier using weak classifiers, our method focuses on filtering the data. A single CNN can achieve high accuracy but low precision and recall for MA detection. However we observe that the performance of the CNN can be enhanced using more balanced data. The aim of MS-CNN is not to ensemble several weak classifiers, but to filter unbalanced data. Therefore the purpose and hypotheses of MS-CNN and boosting are very different.

For the second challenge, the majority of existing work in the context of fundus image processing mainly focus on the green channel extraction as the green channel contains the highest contrast among all the channels. It is believed that the red channel is saturated and the blue channel contains little information, as discussed in [42]. However, since our MS-CNN classifier can handle high dimensional inputs with multiple channels as its input, all three color channels in fundus images, and the lesion map from the image-to-text mapping model as the 4th channels are fed into MS-CNN. We observe that the information in the red and blue channels, though deliver little information in the morphological or wavelet-based method, actually help to improve the classification performance of our proposed MS-CNN model.

IV. EXPERIMENTAL RESULTS

In this section, extensive experiments are carried out to evaluate the clinical report guided multi-sieving CNN model for MA detection in the unbalanced fundus image data. We collect a dataset from the local No. Six People's Hospital. The data set contains fundus images and clinical reports to train and test image-to-text mapping model. We also use the standard diabetic retinopathy (DIARETDB1 [43]) database to test our method.

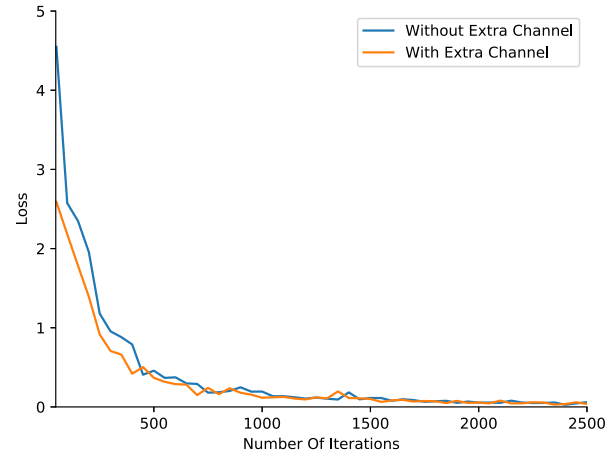


Fig. 5. The converge process of MS-CNN with / without the extra channel. The loss of the network decreases much faster when using the extra channel, which means extra channel brings complementary information to the fundus images.

The dataset collected from the local hospital contains 646 images with text reports. All images were captured using a 45-degree field-of-view digital fundus camera with the resolution of 3504×2336 . 433 of them are without obvious DR and the rest images contain different lesions including hard exudates, soft exudates, microaneurysms, and hemorrhages. Two experts from the hospital annotated MAs in each image. For efficiency, they work separately on the disjoint subsets of all the images collected. In our dataset, each fundus image is captured with a fluorescence fundus angiography image taken at the same time. The two experts annotate fundus images using fluorescence fundus angiography images as reference. Because MAs are clearer and easier to identify in fluorescence images [44], we regard the annotations based on fluorescence images as ground truth.

A. Datasets

DIARETDB1 dataset contains 89 non-dilated fundus images with different kinds of lesions. 84 images contain at least mild non-proliferative signs (MAs) of the diabetic retinopathy, and 5 are considered as normal which do not contain any signs of the diabetic retinopathy according to all experts who participated in the evaluation [43]. Variations in the imaging conditions such as saturation, lighting changes, blurs, and reflected light can be observed in the dataset, which makes the detection task a real challenge. Four ophthalmologists annotated all images independently. Since the annotations of MAs have high inter-observer variance, the union of annotated MAs from all observers is used as the ground truth (i.e. an area is regarded as MA if at least one annotator marks it as MA).

B. Image-To-Text Mapping

We first evaluated the image-to-text mapping algorithm. In our implementation, we used 1000 random ferns, and each fern has eight random functions. That is, we divide the feature space into 256 parts, and perform the partition 1000 times. In the "Mapping Visual Features to Lesion Types"

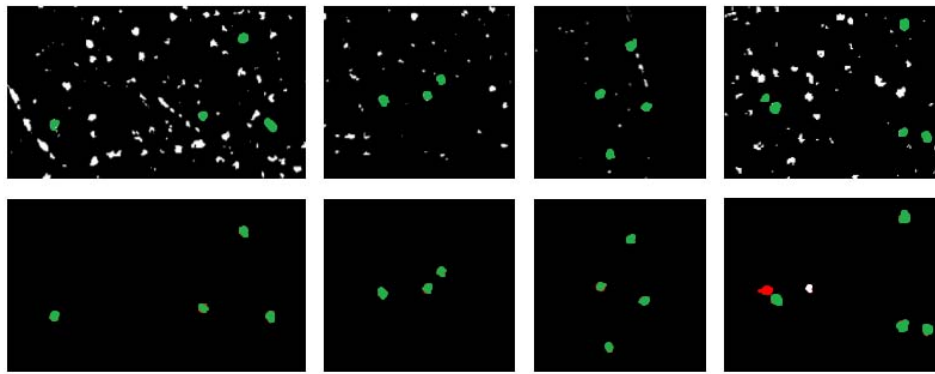


Fig. 6. Illustration of segmentation results of the first and second block in MS-CNN. White, red, and green dots represent false positive, false negative, true positive predictions respectively.

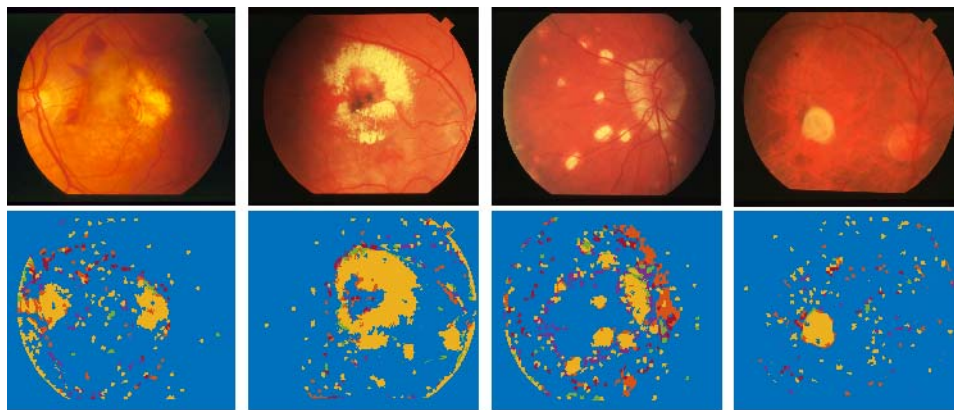


Fig. 7. Illustration of expert knowledge learning results. The top row is the original fundus images with different kinds of lesions. The bottom row is the corresponding output of expert knowledge model. The original output images are gray level images where different gray levels represent different lesion type. We transformed the original output images to pseudo-color images for visualization purpose.

TABLE II
CLASSIFICATION PERFORMANCE OF DIFFERENT LESIONS TYPES
USING THE IMAGE-TO-TEXT MAPPING MODEL

Lesion Type	Accuracy	Recall	Precision	F1
Microaneurysms	0.997	0.141	0.244	0.179
Hard Exudates	0.981	0.912	0.849	0.879
Soft Exudates	0.984	0.778	0.701	0.737
Hemorrhage	0.984	0.581	0.471	0.520

step, we set the number of top partitions to 50. The number of ferns, the number of random functions and the number of top partitions are selected empirically. Since only the dataset from the local hospital is provided with clinical reports, we evaluate the algorithm on this local dataset. 10% of the images are randomly selected from 646 images as the test set and the rest images are used for training. The experiment results are shown in Table II. From the table, we can see that all lesion types have high accuracy but MA and hemorrhage have low F1 scores, which is caused by the unbalanced dataset. Since the number of pixels with lesions are much smaller than the number of pixels in the healthy issues of the fundus images (ratio around 1:4000), simply classifying all pixels as healthy tissue will yield a high accuracy exceeding 97.5%. We observe that recall, precision, and F1 score of MA detection are low.

This indicates that the image-to-text mapping algorithm cannot handle such small and inapparent lesion types by itself. The lesion sizes of exudates and hemorrhages are much larger than microaneurysms. We also observe that the classification scores of exudates and hemorrhages are much higher than MAs. Hard exudates, the most obvious lesion among all lesions, can be well detected with recall of 0.912 and precision of 0.84. The image-to-text mapping algorithm provides the possibility to leverage the wealth of domain knowledge embedded in the clinical reports which are usually left out in the current lesion detection methods for fundus images.

The loss of networks with or without the extra channel are plotted in Fig. 5. It is obvious that the converging speed of the MS-CNN with the extra channel is much faster than the MS-CNN without the extra channel. Two networks are trained using the same parameters. This indicates that MS-CNN is easier to train and converge if the extra image-to-text mapping information is introduced. It further indicates that the clinical reports contain complementary information to the visual features in the fundus images.

C. Multi-Sieving CNN

In our proposed multi-sieving CNN model, AlexNet [30] is used in our experiments as the basic CNN architecture.

TABLE III

EXPERIMENTAL RESULTS. WE COMPARE WITH SEVERAL STATE-OF-THE-ART METHODS FOR MA DETECTION IN FUNDUS IMAGES. WE ALSO COMPARE OUR PROPOSED FRAMEWORK WITH/WITHOUT CLINICAL REPORT GUIDANCE AND A DIFFERENT NUMBER OF MS-CNN BLOCKS. THE BEST RESULTS ARE HIGHLIGHTED USING BOLD FONT

Methods	Recall	Precision	Accuracy	F1 score
Latim [16]	0.805	0.744	0.805	0.773
Fujita Lab [15]	0.715	0.703	0.713	0.709
C.Aravind [45]	0.800	0.920	0.900	0.856
Fulong Ren [46]	0.821	0.961	0.962	0.886
Eftal Sehirli [47]	0.691	0.993	-	0.815
FCN segmentation [26]	0.149	0.9655	0.722	0.242
MS-CNN(Second layer only)	0.034	0.996	0.321	0.466
MS-CNN w/o clinical reports	0.842	0.988	0.951	0.909
1st stage of 2-stage MS-CNN	0.179	1.000	0.178	0.304
2-stage MS-CNN	0.878	0.997	0.961	0.934

We evaluate the efficacy of our proposed framework in terms of the classification accuracy and computational efficiency. There are around 3270 super-pixels in each fundus image, and in total 1016 MAs in the DIARETDB1 dataset.

All of our experiments were performed using the CNN with the same configuration as AlexNet [48]. We train and test the image-to-text mapping model using dataset collected from the local hospital. 80% of images are randomly selected as the training set and the rest are the testing set. Because text reports are not available in DIARETDB1, we use the same image-to-text mapping model trained using the local dataset to perform tests on DIARETDB1 dataset. Following the experiment setup in [43], 28 images in the dataset are pre-selected as the training data and the rest images are used as the testing data. To show that the extra channel helps the MA detection, we train and test two CNNs with the same architecture, but one with the extra expert knowledge guiding channel, one with the extra channel filled with zeros to avoid CNN structural change.

Table III shows the comparison between our proposed method and the state-of-the-art methods. We have implemented two existing methods [15], [16], both of which are designed to process low-resolution images. The parameters in these methods can be either tuned by grid search (e.g. thresholds) or adjusted by normalization (e.g. image size and average intensity), to achieve the optimal combinations. The evaluation results on the DIARETDB1 data set are comparable to the results reported in the original papers. We also compared our results with the reported results of three existing methods published recently [45]–[47]. Our method achieved the highest score in recall and precision compared to all other methods, with comparable accuracy to [46]. We also observe from Table III that without the clinical report guided information, the proposed MS-CNN method already outperforms the state-of-the-art methods in precision, recall and F1 score with a large margin. Furthermore, with the clinical report guidance, the proposed method remarkably boost recall by 5.7% compared with the best of the five methods [46]. We also observe that there is a significant increase of recall from 84.2% to 87.8% when we add the image-to-text mapping channel. This is critical to medical image analysis, as false negatives are considered to be detrimental for the disease diagnosis. This notable improvement in recall indicates that while most

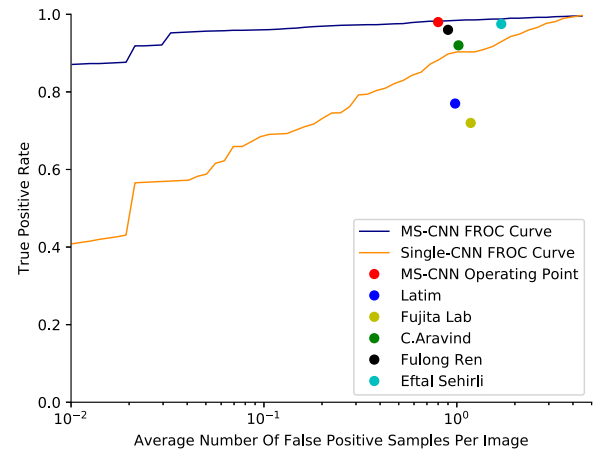


Fig. 8. The FROC curve of MS-CNN model and the operating points of existing methods. The operating points of filtering-based methods Latim [16], Fujita Lab [15] are far below the FROC curve of our proposed method and other learning-based methods (C.Aravind [45], Fulong Ren [46], and Eftal Sehirli [47]). While Eftal Sehirli [47]'s method is closest to our FROC curve, our method generates the fewer false positive predictions than Eftal Sehirli [47]'s method with the same true positive rate.

MAs are too vague to be distinguished from the background, our image-to-text mapping model can effectively fill in the semantic gap between the clinical reports and the medical images with complementary cues.

On the other hand, the minor improvement of precision from 98.8% to 99.7% with the additional clinical report information implies that our proposed multi-sieving scheme is able to eliminate most of the false positive predictions even without the extra information from clinical reports but can further improve the precision with this extra information. Furthermore, we also experiment with the number of blocks in MS-CNN and find that the performance gain is negligible after two blocks while the computational cost increases linearly with the number of blocks. Thus, all of our experiments use two blocks in the MS-CNN framework.

Fig. 8 shows the Free-Response Receiver Operating Characteristic (FROC) curve of the proposed MS-CNN model. Fig. 8 clearly shows that the operating points of previous filtering-based methods [15], [16] are significantly lower than the learning-based methods [45]–[47]. We evaluate the performance of the three learning-based methods [45]–[47] and

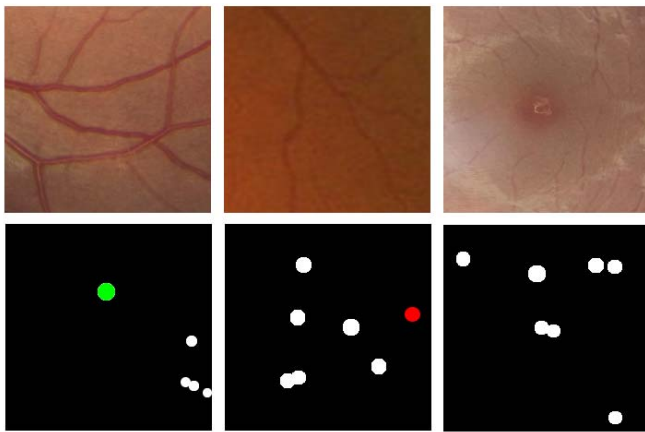


Fig. 9. Illustration of segmentation result of using only Block 2 of the 2-stage MS-CNN. The top row shows original images, and the second row shows the detection results. White dots, red dots and green dots represent false positive, false negative, true positive predictions respectively.

the FROC curve of our MS-CNN model. We observe that the performances of these three methods are inferior to MS-CNN. Even compared with Fulong Ren's method [46] whose performance almost falls on the FROC curve of MS-CNN, MS-CNN can still generate significantly fewer false positive samples in MAs detection.

To further demonstrate that multi-sieving scheme is effective, we also extract and visualize the detection results at the first and second block of MS-CNN. From the last two rows of Table III, we observe that recall increases sharply from 17.9% to 87.7%. This observation shows that though the extra information from the clinical reports could lead to a sharp increase of FP initially when it is coupled with an appropriate number of multi-sieving CNN blocks, the extra expert knowledge would help to boost the overall performance of the final classifier, as shown in Fig. 6. But we also noticed that precision decreases slightly from 100% to 99.7%, as we have to trade off between precision and recall. Since false negative is more detrimental than false positive in disease screening, a slight decrease of precision is acceptable given the improved recall and overall performance.

We also experiment with using only the second layer of MS-CNN. Without eliminating most obvious false positive samples, the second layer fails to distinguish vessels, reflection lights, and other red tissues that are similar to MAs. As shown in Fig. 9, vessels conjections, reflections are recognized as MA by second layer without the first layer. While these tissues are successfully eliminated by the first layer.

D. Fully Convolutional Networks

We also compare with fully convolutional networks (FCN) [26] for MA detection to explore if the approaches in semantic segmentation can be directly applied to this use case.

FCNs have no fully-connected layers, which are typically used for classification. Instead, FCNs use convolutional layers to classify each pixel in the image. So the

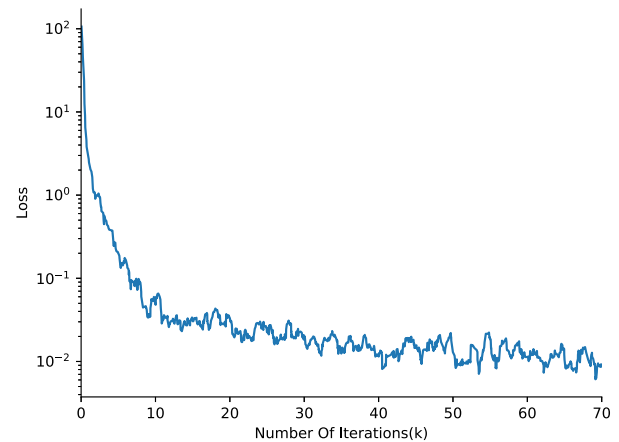


Fig. 10. The training loss of FCN. After fine tuning FCN, the training loss drops dramatically and converges around 70k iteration.

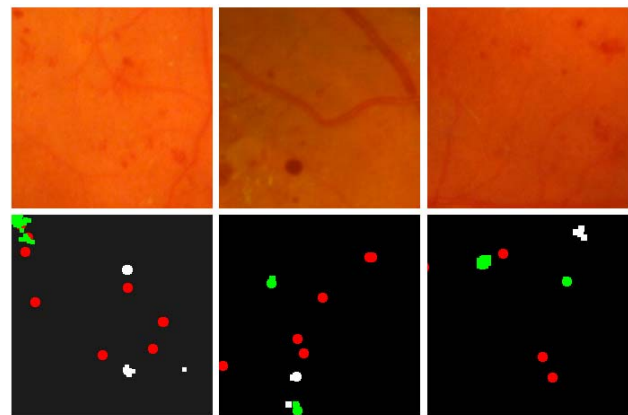


Fig. 11. Illustration of segmentation result of FCN. White dots, red dots and green dots represent false positive, false negative, true positive predictions respectively. It is obvious that there are a lot of false negative samples.

final output layer will be the same height and width as the input image. An FCN is trained end-to-end, pixel-to-pixel. Its performance on semantic segmentation exceeds state-of-the-art methods without further processing or tuning. In this experiment, we train an FCN using a pre-trained model from MatConvNet [49]. To accelerate the training phase, we randomly sample the original image into 229×229 patches. The original image patches are augmented as described in [26]. After training and fine-tuning, the loss of FCN drops down from 400 to 0.01 as shown in Fig. 10. But the testing result is very disappointing, as shown in Fig. 11. MAs are very small even in image patches, and the FCN tends to classify MAs to none-MAs and has a high false-positive rate. To prevent the negative impact of the unbalanced problem, we try to balance the positive and negative samples by reducing the patch size to 100×100 and oversampling the patches with MAs. However, there is no significant improvement observed either.

E. Computational Complexity

We perform all experiments using Intel Xeon E5-2630 v4 @ 2.20GHz CPU and NVIDIA Tesla K80 GPU on Ubuntu 16.04.

We implement the Image-to-Text Mapping component in Python and the Candidate Selection component in C++. We use Caffe [50] implementation of Alex-Net to get the final classification result. In the Image-to-Text Mapping phase, it takes 23 seconds per image to generate the extra channel, where super segmentation takes around 10 seconds and features extraction from the super-pixels takes 12 seconds. The mapping using random ferns takes less than 1 second per image. In the MA detection component, another 28 seconds are spent to generate candidates of MAs per image. Lastly, the time it takes to classify candidates varies from less than 5 seconds to 30 seconds due to the varying number of candidates in each image. It takes 26 seconds on average to generate the final detection results. Overall it takes 77 seconds per image to detect MAs in fundus images. In the super-segmentation, feature generation, and the classification components, the computational efficiency can be further improved using multi-processors since this task can be well paralleled.

The high computing burden is mostly in the training stage and can be alleviated by parallelization. For instance, in the image-to-text mapping phase, most of the time is spent on feature extraction using a CNN. This task can be efficiently paralleled, which means it can be easily distributed on multiple machines and reduce the response time. The super-pixel segmentation also takes relatively long time because it is an iterate approach and cannot be paralleled. Greedy approximation algorithms can be developed to accelerate this procedure. In the testing stage, since we use the patch-wise approach to locate MAs, there can be 100 to 3000 patches of MA candidates for classification. The classification of each phase is independent and can be accelerated by parallelization. Besides, the candidate selection algorithms can be improved to produce less positive candidates. But it is a trade-off between the complexity of candidate selection and classification accuracy.

F. Discussion

Our proposed method has two major advantages. First, we extract expert knowledge from clinical text reports and learn a statistical model for image-to-text mapping. As annotating lesions is costly, learning from the weakly supervised data helps to reduce the need for large-scale labeled training data in deep learning. The image-to-text mapping model is also a general model for learning weak labels using natural language processing. It can benefit a broad range of clinical pathology detection and natural image/video understanding with text information. Second, the MS-CNN algorithm is a generic framework for highly unbalanced data. The key idea is to use multiple classifiers to construct new training data with hard samples. Each block in the MS-CNN structure can be seen as a filter that eliminates the easy samples. As the higher blocks have fewer but more challenging training data as input, they can focus on classifying the harder samples. In fact, each block can adopt a different classifier, which increases the flexibility for tackling practical problems of different nature.

The limitation of MS-CNN is efficiency. Due to the cascaded structure of the MS-CNN model, the training time complexity is $O(N \times M)$, where N is the number of training

examples and M is the number of blocks in MS-CNN. The testing time complexity is $O(M)$ where M is the number of blocks.

V. CONCLUSION

This paper presents a novel clinical report guided framework for automatic MA detection from fundus images. We first extract expert knowledge from clinical text reports through text mining techniques and map visual features to semantic profiles. Integration of keywords information from text reports and fundus images help to boost the detection accuracy with promising performance in terms of precision and recall. The proposed framework performs favorably by overcoming MA detection challenges including unbalanced dataset, varying imaging conditions, and misleading visual cues faced by the existing approaches, via multi-modality information integration from clinical reports and visual features, and multi-sieving classification strategy. The framework proposed in this paper is a generic approach which can be easily extended to detection of multiple kinds of lesions in fundus images and other medical imaging modalities.

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